

Oppgave 1 La δ være Diracs deltafunksjon. Løs initialverdiproblemet

$$y''(t) + y(t) = \delta(t - \pi) + \delta(t - 2\pi) \quad y(0) = y'(0) = 0$$

og skisser løsningen.

$$\begin{aligned} f(t) &= F(s) \\ f'(t) &= y'(t) \Rightarrow F(s) = sF(s) - f(0) \\ f''(t) &= y''(t) \Rightarrow F(s) = s^2 F(s) - sf(0) - f'(0) \end{aligned}$$

$$y''(t) + y(t) = \delta(t - \pi) + \delta(t - 2\pi), \quad y(0) = y'(0) = 0$$

$$F''(s) + F(s) = \delta(s - \pi) + \delta(s - 2\pi)$$

$$s^2 F(s) - sF(0) - F'(0) + F(s) = e^{-\pi s} + e^{-2\pi s}$$

$$F(s)(s^2 + 1) = e^{-\pi s} + e^{-2\pi s}$$

$$F(s) = \frac{e^{-\pi s} + e^{-2\pi s}}{(s^2 + 1)} = \frac{e^{-\pi s}}{(s^2 + 1)} + \frac{e^{-2\pi s}}{(s^2 + 1)}$$

↙ invers Fourier

$$f(t) = f(t - a) u(t - a) \Leftrightarrow e^{-sa} F(s) = F(s)$$

$$f(t) = u(t - \pi) \sin(t - \pi) + u(t - 2\pi) \sin(t - 2\pi)$$

Oppgave 2 La f være en funksjon, og anta at det er mulig å skrive

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}$$

på intervallet $[-\pi, \pi]$.

a) Vis at

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx.$$

b) La $f(x) = e^{-|x|}$. Finn fourierrekken til f på $[-\pi, \pi]$, og skisser denne på intervallet $[-3\pi, 3\pi]$.

$$a) \quad f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \quad c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \quad / \cdot e^{-inx}$$

$$f(x) e^{-inx} = \sum_{n=-\infty}^{\infty} c_n e^{inx} e^{-inx}$$

$$\int_{-\pi}^{\pi} f(x) e^{-inx} dx = \sum_{n=-\infty}^{\infty} c_n \int_{-\pi}^{\pi} 1 dx =$$

$$\int_{-\pi}^{\pi} f(x) e^{-inx} dx = 2\pi c_n \Rightarrow c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \quad \square$$

$$b) \quad f(x) = e^{-|x|}, \quad x \in [-\pi, \pi]$$

$$f(x) \sim \sum_{n=-\infty}^{\infty} c_n e^{inx}$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-|x|} e^{-inx} dx$$

$$= \frac{1}{2\pi} \left(\int_{-\pi}^0 e^x e^{-inx} dx + \int_0^{\pi} e^{-x} e^{-inx} dx \right)$$

$$= \frac{1}{2\pi} \left(\int_{-\pi}^0 e^{x(1-in)} dx + \int_0^{\pi} e^{-x(1+in)} dx \right)$$

$$= \frac{1}{2\pi} \left(\left[\frac{e^{x(1-in)}}{1-in} \right]_{-\pi}^0 - \left[\frac{e^{-x(1+in)}}{1+in} \right]_0^{\pi} \right)$$

$$= \frac{1}{2\pi} \left(\left(\frac{1}{1-in} - \frac{e^{-\pi(1-in)}}{1-in} \right) - \left(\frac{e^{-\pi(1+in)}}{1+in} - \frac{1}{1+in} \right) \right)$$

$$= \frac{1}{2\pi} \left(\frac{1 - e^{-\pi}(-1)^n}{1-in} - \frac{e^{-\pi}(-1)^n + 1}{1+in} \right)$$

$$= \frac{1 - e^{-\alpha}(-1)^n}{2\alpha} \left(\frac{1}{1-in} + \frac{1}{1+in} \right), \quad \frac{1}{1-in} = \frac{1+in}{1+n^2}, \quad \frac{1}{1+in} = \frac{1-in}{1+n^2}$$

$$= \frac{1 - e^{-\alpha}(-1)^n}{2\alpha} \left(\frac{2}{1+n^2} \right)$$

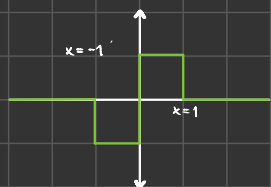
$$= \frac{1 - e^{-\alpha}(-1)^n}{\alpha(1+n^2)}$$

$$f(x) \sim \sum_{n=-\infty}^{\infty} c_n e^{inx} = \frac{1-e^{-\alpha}}{\alpha} + \sum_{n \neq 0} \frac{1 - e^{-\alpha}(-1)^n}{\alpha(1+n^2)} e^{inx}$$

Oppgave 3 La u være heavisidefunksjonen

$$u(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$$

Skisser funksjonen $-u(x+1) + 2u(x) - u(x-1)$, og finn dennes fouriertransform.



$$u(x) = \begin{cases} -1, & -1 \leq x < 0 \\ 1, & 0 \leq x < 1 \\ 0, & \text{ellers} \end{cases}$$

$$F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx = \frac{1}{\sqrt{2\pi}} \left(\int_{-1}^0 e^{-i\omega x} dx + \int_0^1 e^{-i\omega x} dx \right) = \frac{1}{\sqrt{2\pi}} \left(\left[\frac{e^{-i\omega x}}{-i\omega} \right]_{-1}^0 - \left[\frac{e^{-i\omega x}}{-i\omega} \right]_0^1 \right)$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{1 - e^{-i\omega}}{-i\omega} - \left(\frac{e^{-i\omega} - 1}{-i\omega} \right) \right) = \frac{1}{\sqrt{2\pi} i\omega} (1 - e^{-i\omega} - e^{-i\omega} + 1) = \frac{1}{\sqrt{2\pi} i\omega} (2 - (e^{i\omega} + e^{-i\omega})) = \frac{2 - 2\cos(\omega)}{\sqrt{2\pi} i\omega} = \frac{\sqrt{2}(1 - \cos(\omega))}{\sqrt{\pi} i\omega}$$

Oppgave 4 Finn løsningen til varmelikningen

$$u_t = u_{xx},$$

for $0 \leq x \leq \pi$ og $t \geq 0$ med randkrav

$$u_x(0, t) = u_x(\pi, t) = 0$$

og initialkrav

$$u(x, 0) = e^{-x}.$$

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-\left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi}{L} x\right), \quad A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L} x\right) dx$$

$$A_n = \frac{2}{\pi} \int_0^{\pi} e^{-x} \sin(nx) dx \quad \int u v' = uv - \int u' v, \quad u = e^{-x}, u' = -e^{-x}, v = -\frac{\cos(nx)}{n}, v' = \sin(nx)$$

$$\int_0^{\pi} e^{-x} \sin(nx) dx = -\frac{e^{-x} \cos(nx)}{n} - \frac{1}{n} \int_0^{\pi} e^{-x} \cos(nx) dx \quad u = e^{-x}, u' = -e^{-x}, v = \frac{\sin(nx)}{n}, v' = \cos(nx)$$

$$= -\frac{e^{-x} \cos(nx)}{n} - \frac{1}{n} \left(e^{-x} \frac{\sin(nx)}{n} + \frac{1}{n} \int_0^{\pi} e^{-x} \sin(nx) dx \right)$$

$$\int_0^{\pi} e^{-x} \sin(nx) dx = -\frac{e^{-x} \cos(nx)}{n} - \frac{e^{-x} \sin(nx)}{n^2} - \frac{1}{n^2} \int_0^{\pi} e^{-x} \sin(nx) dx$$

$$\int_0^{\pi} e^{-x} \sin(nx) dx + \frac{1}{n^2} \int_0^{\pi} e^{-x} \sin(nx) dx =$$

$$\int_0^{\pi} e^{-x} \sin(nx) dx \left(1 + \frac{1}{n^2} \right) = -\frac{e^{-x}}{n^2} (n \cos(nx) + \sin(nx))$$

$$\int_0^{\pi} e^{-x} \sin(nx) dx = -\frac{e^{-x} (n \cos(nx) + \sin(nx))}{n^2 + 1} \Big|_0^{\pi} = \frac{n(1 - e^{-\pi}(-1)^n)}{n^2 + 1} \Rightarrow A_n = \frac{2n(1 - e^{-\pi}(-1)^n)}{\pi(n^2 + 1)}$$

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-\left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi}{L} x\right) = \sum_{n=1}^{\infty} \frac{2n(1 - e^{-\pi}(-1)^n)}{\pi(n^2 + 1)} e^{-n^2 t} \sin(nx)$$

Oppgave 5 Anta at du har en tabell med $n+1$ forskjellige punkter på x -aksen, og tilhørende funksjonsverdier. Vis at det finnes et entydig polynom av maksimal grad n som interpolerer tabellen.

Formlene som blir brukt for å interpolere tabellen multipliserer n av $n+1$ verdier

sammen, altså får vi et polynom av gr. n .

Oppgave 6 La

$$f(x) = \cos x.$$

Bruk en endelig differanseformel til å tilnærme $f'(1) \approx -0.841470984807897$ med en feil på mindre enn 10^{-9} .

$$f(x+k) = \sum \frac{f^{(n)}(x)}{n!} k^n = f(x) + f'(x)k$$

$$f(x-k) = \sum \frac{f^{(n)}(x)}{n!} (-k)^n = f(x) - f'(x)k$$

$$\text{Sentral differanse: } f(x+k) - f(x-k) = f(x) + f'(x)k - (f(x) - f'(x)k) \Rightarrow f'(x) = \frac{f(x+k) - f(x-k)}{2k}$$

$$f'(1) = \frac{f(1+k) - f(1-k)}{2k} \leq -0.841470984, \quad k=0.001 \quad \text{10K?}$$

Oppgave 7 Finn en tilnærming til integralet

$$\int_{-1}^1 x^8 + x^6 dx = \frac{32}{63}$$

ved hjelp av fempunkts Gauss-Legendre-kvadratur.

HUSK! $n+1$ -pkts G-L-regel er eksakt for poly. av gr. $2n+1$

$$n+1=5 \Rightarrow n=4, \quad 2n+1=9 \Rightarrow \sum_{i=1}^4 A_i f_i$$

$$\sum_{i=1}^4 A_i f_i = \omega_1 f(x_1) + \omega_2 f(x_2) + \omega_3 f(x_3) + \omega_4 f(x_4)$$

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